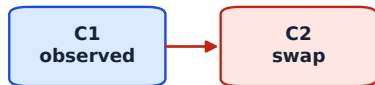


FairEval experimental geometry

Each RQ reuses the same frozen recommendation task and changes only the factor required by its registered estimand.

RQ1 Demographic consequence

MATCHED CONTRAST



$$\text{CUG} = U(C1) - U(C2)$$

Scope
MovieLens-1M + Last.fm-1K

gender confirmatory • age robustness
RBO / Jaccard = sensitivity only

Inference
paired sign-flip + bootstrap CI
user is the unit

RQ2 Grounded personality value

MEASURED CONTROL



$$\text{PVA} = U(C3) - U(C4)$$

Scope
3 measured-personality datasets

whole-profile derangement • C5 one-trait
dataset-native psychometrics only

Inference
same paired user-level stack
true vs shuffled is primary

RQ3 Reliability + generalization

REGISTERED FACTORS

Prompt	3 paraphrases
Cue	3 realizations
Order	3 permutations
K	5 / 10 / 20
Repeat	3 main; 10 stress

FairSynth-360 sanity layer
A/B/C identity has no relevance effect
real / synthetic reported separately

No best-prompt selection

RQ4 Personalization-preserving mitigation

MITIGATION FRONTIER



Validation-only selection
≥95% baseline validation nDCG
then minimize |CUG|
freeze (α, λ) before test

No test-set point selection
report “no eligible point” if needed

Goal
reduce harmful gap without
erasing useful personalization

Shared contract • frozen prompt + candidates • repeated generations • invalid-output accounting • paired user-level inference • Holm correction